

for typos, copy and paste it to the online diff tool at <https://inventwithpython.com/beyond/diff/>.

## Writing the Code

Let's look at the program's source code, as we did for the Tower of Hanoi program. Once again, I formatted this code using Black with a line limit of 75 characters.

We'll begin at the top of the program:

---

```
"""Four-in-a-Row, by Al Sweigart al@inventwithpython.com
A tile-dropping game to get four-in-a-row, similar to Connect Four."""

import sys

# Constants used for displaying the board:
EMPTY_SPACE = "." # A period is easier to count than a space.
PLAYER_X = "X"
PLAYER_O = "O"
```

---

We start the program with a docstring, module imports, and constant assignments, as we did in the Tower of Hanoi program. We define the `PLAYER_X` and `PLAYER_O` constants so we don't have to use the strings "X" and "O" throughout the program, making errors easier to catch. If we enter a typo while using the constants, such as `PLAYER_XX`, Python will raise `NameError`, instantly pointing out the problem. But if we make a typo with the "X" character, such as "XX" or "Z", the resulting bug might not be immediately obvious. As explained in “Magic Numbers” on page 71, using constants instead of the string value directly provides not just a description, but also an early warning for any typos in your source code.

Constants shouldn't change while the program runs. But the programmer can update their values in future versions of the program. For this reason, we make a note telling programmers that they should update the `BOARD_TEMPLATE` and `COLUMN_LABELS` constants, described later, if they change the value of `BOARD_WIDTH`:

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```
# Note: Update BOARD_TEMPLATE & COLUMN_LABELS if BOARD_WIDTH is changed.
BOARD_WIDTH = 7
BOARD_HEIGHT = 6
```

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